

StaySense BI

Week 6 – AI Validation Log

Project: StaySense BI – AI-Powered Guest Sentiment Intelligence for Homestays

Dataset: Cleaned NYC Airbnb 2019 Dataset

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AI Prompt Used

I have a cleaned Airbnb dataset containing information such as borough, room type, price, reviews, availability, and minimum nights. The dataset has already been used to build a Business Intelligence dashboard. Suggest five business questions that would help Airbnb hosts or homestay owners make better business decisions using this data.

AI-Generated Questions

AI Question	Evaluation	Reason
Which borough has the highest average listing price?	✅ Accurate	Directly answerable using borough and price columns.
Which room type attracts the highest customer engagement based on reviews?	✅ Accurate	Review counts are available for every room type.
Is there a relationship between availability and customer reviews?	⚪ Partially Accurate	The data supports analysis, but the scatter plot shows no strong linear relationship.
Which price category contains the largest number of listings?	✅ Accurate	Price categories were created and analysed successfully.
Which borough provides the best balance between affordability and guest popularity?	⚪ Partially Accurate	Requires combining multiple metrics and business interpretation rather than a direct calculation.

AI Claims Edited

Example 1

AI Draft: Manhattan is the best borough for every Airbnb investment.

Edited Version: Manhattan has the highest average listing price in this dataset, indicating strong revenue potential. However, investment decisions should also consider occupancy, competition, and operating costs.

Reason: The AI overgeneralized by claiming Manhattan is always the best investment. The dataset supports higher pricing but not overall investment performance.

Example 2

AI Draft: Listings with higher availability receive more customer reviews.

Edited Version: The analysis shows no strong linear relationship between availability and customer reviews. While some highly available listings receive many reviews, many others do not.

Reason: The scatter plot did not support a strong positive correlation.

Example 3

AI Draft: Entire home listings are preferred by all guests.

Edited Version: Entire home/apartment listings accumulated the highest total number of reviews, indicating higher overall guest engagement within this dataset.

Reason: The AI made an absolute claim ("all guests"), while the data only supports higher engagement, not universal preference.

Conclusion

AI-assisted analysis significantly accelerated the generation of business questions and initial interpretations. However, every insight was verified using the actual dataset. Several AI-generated statements required refinement to ensure they accurately reflected the evidence. This demonstrates that AI is an effective analytical assistant but should complement, rather than replace, data-driven decision-making.